

Alidade wire harnessing Enclosure								
Reference ELMCOview.vsd								
24-Mar-06								
AWE1 Input power wires through pedestal Belden 8527/8520								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC			white	#12		
	2	Return			black	#12		
	3	Ground			green	#14		
Ends	A	bare wire		screw terminal block behind pie-plate				
	B	39860-0103		Molex terminal on Alidade Relay Board				
AWE2 Transformer								
	A	Description		Comment	Color	Size	Twist	Length
Wires	1	120VAC		These wires except Ground are all on the toroid, we provide connector	white	#14		
	2	Return			black	#14		
	3							
	4	100VAC			red	#14		
	5	Return			red	#14		
End	A	39860-0105		Molex connector on alidade relay card				
AWE3 Power to motors								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	110VAC		Use standard power cord and chop off plug	black	#14		2 ft
	2	Return			white	#14		
	3	Ground			green	#14		
Ends	A	39860-0104		Molex connector on alidade relay board				
	B	IEC-320		plug on drive box				
AWE4 Power for control box								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC		Use standard power cord and chop off plug	white	#18		
	2	Return			black	#18		
	3	Ground			green	#18		
Ends	A	39860-0103		Molex connector on alidade relay board				
	B	IEC-320		plug on drive box				
AWE5 Power for rim power box (also called AR1) wire								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	120VAC			black	#14		
	2	Return			white	#14		
	3	Ground			green	#14		
Ends	A	39860-0103		Molex connector on alidade relay board				
	B	ParlBlade U Gnd		connector in rim power box				
AWE6 15-wire cable from alidade relay board to drive box CablesToGo custom								
	A	Description	B	Comment		Size		Length
Wires/pins	1	az brakes +24V	1	1&9 are controlled with the main drives transformer power				
	2	Ground	2					
	3	24V aux	3			all #24		23"
	4	Ground	4					
	5	drive enable +	5	5&13 are connected through relay contacts which are closed when the main drives transformer power relay is latched				
	6	Ground	6					
	7	Az cw limit	7	7&15 are connected to 8 via NC switches				
	8	Az limit common	8					
	9	el brakes +24V	9					
	10	Ground	10	Belden 9947 maybe				
	11	24V aux	11	awaiting quote from cablestogo.com				
	12	Ground	12					
	13	drive enable -	13					
	14	Ground	14					
	15	Az ccw limit	15					
Ends	A	D15P		Dsub to alidade relay board				
	B	D15S		Dsub to drive box	connects to AWD2A			

AWE7	25-wire cable from alidade relay board to control box				CablesToGo custom			
	A	Description	B	Comment				Length
Wires/pins	1	brakes +24V	1	1,3,5,11,14,16,18, 24 are power and control signals from the				
	2	Ground	2	control box				
	3	24V aux	3					
	4	Ground	4					
	5	5V sensor +	5					
	6	Ground	6					
	7	float in	7	7,8,9,10,12,13,20,21,22,23,25 are logic and sensor signals				
	8	T input1 -	8	returning to the control box				
	9	T input2 -	9					
	10	Az brake sense	10					
	11	Latch	11	main drives transformer power is latched by holding 24 logic high				
	12	Pot 1	12	while pulsing 11 logic low-high-low				
	13	Pot wiper	13	wrap pot is wired to relay board such that resistance between 13				
	14	brakes +24V	14	and 25 increases when azimuth rotates cw				
	15	Ground	15					
	16	24V aux	16					
	17	Ground	17					
	18	5V sensor -	18					
	19	Ground	19					
	20	T input1 +	20					
	21	T input2 +	21					
	22	electronics reset	22					
	23	El brake sense	23					
	24	Unlatch	24					
	25	Pot 2	25					
Ends	A	D25S	Dsub to alidade relay board					
	B	D25P	Dsub to control box					
AWE8	motor power and brakes for azimuth				part of motor			
	A	Description	B, C	Comment	Color	Size	Twist	Length
Wires/pins	1	W1	BW		purple	#14		
	2	W2						
	14	W3						
	15	W4						
	3	N/C						
	16	N/C						
	4	V1	BV		brown	#14		
	5	V2						
	17	V3						
	18	V4						
	6	N/C						
	19	N/C						
	7	U1	BU		blue	#14		
	8	U1						
	20	U1						
	21	U1						
	9	N/C						
	22	N/C						
	10	G1	BGND		green	#14		
	11	G2						
	23	G3						
	24	G4						
	12	BP	C		blue	#16		
	13	BM	C		blue	#16		
	25	SHIELD	N/C		shield	shield		
Ends	A	D25P	at driver board					
	B	at motor						
	C	39-01-3023-P/39-00-0041-C	wires part of brake, but with molex					

AWE9	encoders and hall's for azimuth				part of motor			
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	N/C		GND on board	N/C			
	2	U			brown			
	3	V			gray			
	4	W			white			
	5	GND			black		V	
	6	X			purple		Z	
	7	B			green		Y	
	8	A			blue		X	
	9	thermal sensor		GPIN5	orange		W	
	10	GND thermal sensor		GND	orang/white		W	
	11	Vcc (+5V)			red		V	
	12	N/C		GND on board	N/C			
	13	/X			purple/blk		Z	
	14	/B			green/blk		Y	
	15	/A			blue/blk		X	
Ends	A	D15P	at driver board					
	B	M						
AWE10	extender for motor power and brakes for elevation				Lapp 7072403			
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	W1	4		BLACK 3			160"
	2	W2						
	14	W3						
	15	W4						
	3	N/C	3		BLACK 2			
	16	N/C						
	4	V1						
	5	V2						
	17	V3						
	18	V4						
	6	N/C	2		BLACK 1			
	19	N/C						
	7	U1						
	8	U1						
	20	U1	5		GREEN/ YELLOW			
	21	U1						
	9	N/C						
	22	N/C						
	10	G1						
	11	G2						
	23	G3			BLACK 5			
	24	G4						
	12	BP						
	13	BM			BLACK 6			
	25	SHIELD						
Ends	A	D25P	plugs into D25S on board					
	B	bare wire	MBTO terminal strip					

AWE11 extender for encoders and hall's for elevation Belden 8307								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	N/C		GND on board				160"
	2	U	2		Brown		T	
	3	V	3	Note: take care to	Black		T	
	4	W	4	get the right black	Blue		U	
	5	GND	5	for twisted pair	Black		V	
	6	X	14	as indicated	Orange		Z	
	7	B	12		White		Y	
	8	A	10		Yellow		X	
	9	thermal sensor	7	GPIN5	Green		W	
	10	GND thermal sensor	8	GND	Black		W	
	11	Vcc (+5V)	6		Red		V	
	12	N/C		GND on board				
	13	/X	15		Black		Z	
	14	/B	13		Black		Y	
	15	/A	11		Black		X	
Ends	A	D15P	Dsub pin on end of elevation sensor extender					
	B	bare wire	HTE terminal strip					
AWE12 encoder for azimuth (P/N: R176H-03600Q-5L10-AT40SP-24MN)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	A			yellow			
	2	B			green			
	3	IND			blue			
	4	N/C			shield			
	5	+V			red			
	6	/A			brown			
	7	/B			orange			
	8	/IND			white			
	9	COMMON			black			
Ends	A	D9P	Dsub pin on end of encoder cable					
	B	--	cable part of encoder					
AWE13 encoder for elevation (P/N: 9220S03600D5L10D99SP04EA)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires/pins	1	A			yellow			
	2	B			green			
	3	IND			blue			
	4	N/C			shield			
	5	+5V			red			
	6	/A			brown			
	7	/B			orange			
	8	/IND			white			
	9	COMMON			black			
Ends	A	D9P	Dsub pin on end of encoder cable					
	B	--	cable part of encoder					
AWE14 battery for control box wire								
	A	Description	B	Comment	Color	Size	Twist	
Wires/pins	1	+12V		1 use 1-#18	red	18		
	2	+12V		1 or 3-#22	red	same wire		
	3	N/C						
	4	GND		2 use 1-#18	black	18		
	5	GND		2 or 3-#22	black	same wire		
	6	+12V		1	red	same wire		
	7	N/C						
	8	N/C						
	9	GND		2	black	same wire		
Ends	A	D9S	at controller box					
	B	Spade	at battery					

AWE15	Signal from relay board to pie-plate			LappUSA (Olflex) 900P Cable, 3 conductor #24, 811442				
	A	Description	B	Comment	Color	Size	Twist	
Wires	1	M-stop	1		white	#24		
	2	reset	2		brown	#24		
	3	GND	3		green	#24		
Ends	A	Molex 43645-0300/43030-009	on alidade relay board (Jameco 338360/337738)					
	B	bare		terminal behind pie plate				
AWE16	signal from az oil float			wire				
	A	Description	B	Comment	Color	Size	Twist	
Wire	1	Float 1			red	#24		
	2	Float 2			red			
Ends	A	Molex 43645-0200/43030-009	on alidade relay board (Jameco 338351/337738)					
	B	--						
AWE17	Signal from temperature sensor to relay board			wire				
	A	Description	B, C	Comment	Color	Size	Twist	
Wires	1	+5V	B1		Red	#24		
	2	+5V	C1		Orange	#24		
	3	T (T1)	B3		Yellow	#24		
	4	T (T2)	C3		Yellow	#24		
	5	GND	B2		Black	#24		
	6	GND	C2		Black	#24		
Ends	A	Molex 43025-0600/43030-009	to alidade relay board					
	B	solder to LM35		Lm-35 Device in TO-220				
	C	solder to LM35		Lm-35 Device in TO-220				
AWE18	Signal from az wrap potentiometer			wire				
	A	Description	B	Comment	Color	Size	Twist	
Wires	1	POT1		pin 12 (AWE7)	green	#24		
	2	POT WIPER		pin 13 (AWE7)	black			
	3	POT 2		pin 25 (AWE7)	yellow			
Ends	A	Molex 43645-0300/43030-009	on alidade relay board					
	B	bare wire	posts on pot					
AWE19	Signal from elevation level sensor and thermal sensor			Belden 9611				
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	+5V	10	assume A has odd/	brown	#24		
	2	+5V	11	even rows	red			160"
	3	T (T3)	12		white			
	4	Level (T4)	13	cable may change	green			
	5	GND	14	to Belden 9940	black			
	6	GND	15		blue			
Ends	A	Molex 43025-0600/43030-009	on relay board					
	B	bare wire	MBTO terminal strip					